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City-scale accessibility of emergency responders operating during flood events

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Abstract. Emergency responders often have to operate and respond to emergency situations during dynamic weather conditions, including floods. This paper demonstrates a novel method using existing tools and datasets to evaluate emergency responder accessibility during flood events within the city of Leicester, UK. Accessibility was quantified using the 8 and 10 min legislative targets for emergency provision for the ambulance and fire and rescue services respectively under "normal" no-flood conditions, as well as flood scenarios of various magnitudes (1 in 20-year, 1 in 100-year and 1 in 1000-year recurrence intervals), with both surface water and fluvial flood conditions considered. Flood restrictions were processed based on previous hydrodynamic inundation modelling undertaken and inputted into a Network Analysis framework as restrictions for surface water and fluvial flood events. Surface water flooding was shown to cause more disruption to emergency responders operating within the city due to its widespread and spatially distributed footprint when compared to fluvial flood events of comparable magnitude. Fire and rescue 10 min accessibility was shown to decrease from 100, 66.5, 39.8 and 26.2 % under the no-flood, 1 in 20-year, 1 in 100-year and 1 in 1000-year surface water flood scenarios respectively. Furthermore, total inaccessibility was shown to increase with flood magnitude from 6.0 % under the 1 in 20-year scenario to 31.0 % under the 1 in 100-year flood scenario. Additionally, the evolution of emergency service accessibility throughout a surface water flood event is outlined, demonstrating the rapid impact on emergency service accessibility within the first 15 min of the surface water flood event, with a reduction in service coverage and overlap being observed for the ambulance service during a 1 in 100-year flood event. The study provides evidence to guide strategic planning for decision makers prior to and during emergency response to flood events at the city scale. It also provides a readily transferable method for exploring the impacts of natural hazards or disruptions in other cities or regions based on historic, scenario-based events or real-time forecasting, if such data are available.

1 Introduction

Floods are one of the most significant natural hazards, affecting 116 million people globally, causing approximately 7000 deaths and damages in the region of USD 7.5 billion annually (UNESCO, 2010). Within the UK, the Environment Agency (2009) estimated that 5 million people (1/12 of the UK population) occupying 2 million properties are currently at risk from coastal, fluvial or surface water (pluvial) flooding. Following the Pitt Review (Pitt, 2008), the Environment Agency produced UK-wide surface water flood hazard maps and identified flood "hotspots" at direct flood risk. Although

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自然灾害与地球系统科学
开放获取

洪水期间开展作业的应急响应人员的城市尺度可达性

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摘要。应急响应人员常需在多变的天气条件下开展作业并应对紧急情况，包括洪水。本文展示了一种新方法，利用现有工具和数据集评估英国莱斯特市洪水期间应急响应人员的可达性。在“正常”无洪水条件下，分别依据救护车服务以及消防与救援服务应急保障的法定 8 分钟和 10 分钟目标量化可达性；同时考察了不同量级的洪水情景（重现期分别为 20 年一遇、100 年一遇和 1000 年一遇），并考虑地表水洪水和河流洪水。根据此前开展的水动力淹没模拟结果处理洪水限制条件，并将其作为地表水洪水和河流洪水事件的限制条件输入网络分析框架。研究表明，与量级相当的河流洪水事件相比，地表水洪水因影响范围广且空间分布分散，对在市内开展作业的应急响应人员造成更大的干扰。结果显示，在无洪水、20 年一遇、100 年一遇和 1000 年一遇地表水洪水情景下，消防与救援服务 10 分钟可达性分别从 100% 降至 66.5%、39.8% 和 26.2%。此外，完全不可达的比例随洪水量级增大而上升，从 20 年一遇情景下的 6.0% 增至 100 年一遇洪水情景下的 31.0%。此外，应急服务还描述了地表水洪水事件期间可达性的变化，显示洪水事件发生后的前 15 分钟内，应急服务可达性迅速受到影响；在 100 年一遇洪水事件中，救护车服务的覆盖范围和重叠区域均有所减少。本研究为决策者在城市尺度上制定洪水事件发生前及应急响应期间的战略规划提供了依据。研究还提供了一种易于迁移的方法，可用于探究其他城市或地区遭受自然灾害或中断的影响，依据可以是历史事件、情景模拟事件或实时预测（如有此类数据）。

1 引言

洪水是最重大的自然灾害之一，全球有 1.16 亿人受其影响，每年约造成 7000 人死亡，并导致该地区约 75 亿美元的损失（UNESCO，2010）。在英国，环境署（2009）估计，目前有 500 万人（占英国人口的 1/12）所居住的 200 万处房产面临沿海洪水、河流洪水或地表水（雨洪）风险。根据《皮特审查报告》（Pitt，2008），环境署制作了英国范围的地表水洪水危险性地图，并确定了直接面临洪水风险的“热点”。尽管

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ambulance stations, and fire and rescue stations. Six fire and rescue stations (Birstall, Western, Southern, Central, Eastern and Wigston) and five ambulance and hospital locations (Goodwood Ambulance Station, Leicester Royal Infirmary, Gorse Hill Ambulance Station, Narborough Ambulance Station and Leicester General Hospital) were identified as points of origin for modelling emergency response zones.

2.2.2 Flooding scenarios

The impact of surface water and fluvial flooding on Leicester's emergency response times for ambulance and fire and rescue were both considered. Existing surface water and fluvial inundation datasets associated with flooding of various magnitudes were obtained directly from the Leicester City Council and Environment Agency respectively. Fluvial and surface water flood events with return periods of 1 in 20, 100 and 1000 years were assessed.

High-resolution (1 m horizontal, ±0.25 m vertical), city-wide surface water inundation depth data derived from a hydrodynamic inundation model (TUFLOW), conducted as part of Leicester's Surface Water Management Plan (2012), were obtained from the Leicester City Council. This resource allowed the extraction of spatially referenced flood depth data at multiple points in time throughout the flood event. The modelling involved applying spatially uniform precipitation associated with specified return periods, namely 1 in 20, 1 in 100 and 1 in 1000 years, calculated for design storm hyetographs of 6 h duration (Fig. 2). Distributed roughness values classified according to Ordnance Survey MasterMap© land uses (e.g. 0.02 for roads, 0.03 for buildings, 0.04 for gardens and/or vegetation) were applied in the modelling process. The modelling included a uniform drainage rate of 12 mm h^{-1} to account for drainage and/or infiltration to natural, permeable surfaces and artificial drainage systems such as sewers and manholes, as recommended by the Environment Agency (2012). Further information on the surface water inundation modelling used in this study can be found in Leicester City Council's Surface Water Management Plan (2012).

Fluvial inundation data for the River Soar and associated tributaries within Leicester were obtained from the Environment Agency. As flood depths were not available for fluvial flooding, flood hazard data were used to derive flood restrictions. Flood hazard is a function of flood depth (m), velocity (m s^{-1}) and potential for entrainment of debris within floodwaters (HR Wallingford, 2006), all factors that could inhibit vehicle passage. The spatially distributed flood hazard data were classified into four categories based on the flood hazard rating calculated using the FD23211 guidance document proposed by HR Wallingford (2006): (i) low – shallow flowing water or deep, standing water, (ii) moderate – dangerous for some with deep, fast-flowing water, (iii) significant – dangerous for most with deep, fast-flowing water and (iv) extreme – deep, fast-flowing water that is dangerous to all.

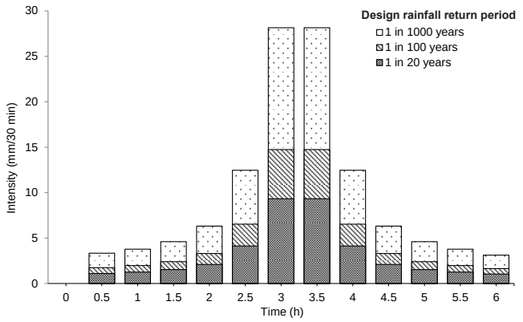


Figure 2. Design rainfall scenarios for the 1 in 20-, 1 in 100- and 1 in 1000-year surface water flood modelling conducted by the Leicester City Council.

2.3 Methods

2.3.1 Network restrictions

First, flood restrictions were defined using the data detailed in the previous section. A study by the Automobile Association (2014) recommended that regular motorists (i.e. small or medium cars) should avoid driving through floodwaters ≥ 15 cm depth as this may be sufficient to stall a car or result in loss of control, while water depths exceeding 30 cm may be sufficient to move vehicles. Additionally, depths ≥ 15 cm may conceal submerged hazards (e.g. overflowing drains or large debris), which could prevent vehicles from successfully traversing floodwaters. Despite this, emergency vehicles have a greater tolerance for travelling through floodwaters than standard vehicles.

Semi-structured interviews conducted with Leicestershire Fire and Rescue Service revealed that water depths of approximately 25 cm (lower than the wheel arch of the vehicle) may be traversed during an emergency situation due to the size, weight and power of emergency vehicles – a fire appliance weighs approximately 12 T and drivers are trained to traverse through floodwaters. This threshold depth value is also consistent with previous research (HR Wallingford, 2006; Dawson et al., 2011; Pregnolato et al., 2016). Although high velocities may hinder emergency responders from successfully traversing floodwaters, modelled flow velocities were typically < 1 m s^{-1} due to ponding in topographic hollows. Therefore, depth was selected as the principal factor when evaluating the sites of network restrictions. Hence, a threshold water depth of 25 cm was set for the surface water flood scenarios, with water depths ≥ 25 cm being treated as restrictions to emergency vehicle flow along a specific road section.

Surface water flood depths ≥ 25 cm were then processed to remove additional polygons that did not overlap or intercept with the ITN and would not be used for analyses (i.e. in

救护站以及消防与救援站。六个消防与救援站（Birstall、Western、Southern、Central、Eastern 和 Wigston）以及五个救护与医院地点（Goodwood 救护站、莱斯特皇家医院、Gorse Hill 救护站、Narborough 救护站和莱斯特总医院）被确定为应急响应区域建模的起点。

2.2.2 洪水情景

研究同时考虑了地表水洪水和河流洪水对莱斯特救护、消防与救援应急响应时间的影响。不同规模洪水相关的现有地表水和河流淹没数据集分别直接取自莱斯特市议会和环境署。评估了重现期为20年一遇、100年一遇和1000年一遇的河流及地表水洪水事件。

从水动力淹没模型（TUFLOW）获得了高分辨率（水平1 m、垂直±0.25 m）的全市地表水淹没深度数据。该模型是莱斯特地表水管理计划（2012）的一部分，由莱斯特市议会实施。该数据资源可提取洪水事件期间多个时间点上具有空间参考的洪水深度数据。建模采用与指定重现期相对应的空间均匀降水，即20年一遇、100年一遇和1000年一遇降水；这些降水根据历时6 h的设计暴雨雨型计算（图2）。按照 Ordnance Survey MasterMap© 分类的分布式粗糙度值

土地利用值（例如道路为0.02、建筑物为0.03、花园和/或植被为0.04）用于建模过程。建模采用统一排水率

12 mm h-1，以计入自然可渗透地表的排水和/或入渗，以及下水道和检查井等人工排水系统的作用，此做法遵循环境署（2012）的建议。关于本研究所用地表水淹没建模的更多信息，见莱斯特市议会《地表水管理计划》（2012）。

莱斯特境内索尔河及其相关支流的河流淹没数据取自环境署。由于没有河流洪水的淹没深度数据，因此使用洪水危险度数据来推导洪水限制。洪水危险度是洪水深度（m）、流速（m s-1）以及洪水中碎屑可能被卷带的程度的函数（HR Wallingford，2006），这些因素都可能阻碍车辆通行。空间分布的洪水危险度数据依据 HR Wallingford（2006）提出的 FD23211 指南文件计算的洪水危险度评级分为四类：（i）低——浅层流动水或深层静水；（ii）中等——深而湍急的水流对部分人构成危险；（iii）显著——深而湍急的水流对大多数人构成危险；（iv）极端——深而湍急、对所有人都构成危险的水流。

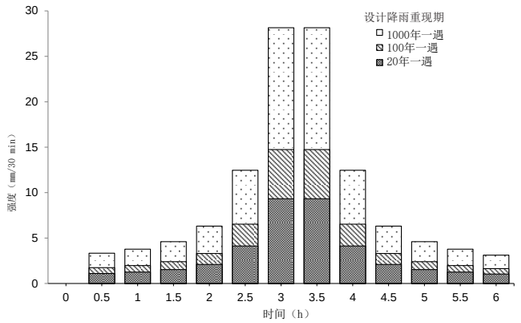


图2。莱斯特市议会开展地表水洪水建模所采用的20年一遇、100年一遇和1000年一遇设计降雨情景。

2.3 方法

2.3.1 网络限制

首先，依据上一节所述数据确定洪水限制。汽车协会（2014）的一项研究建议，普通驾驶者（即小型或中型轿车）应避免驶入水深≥15 cm的洪水，因为这一深度可能足以使汽车熄火或导致失控；而水深超过30 cm时，可能足以推动车辆。此外，水深≥15 cm时，水下危险物（例如溢流排水口或大型碎屑）可能被遮蔽，从而阻碍车辆成功穿越洪水。尽管如此，应急车辆穿越洪水的能力比普通车辆更强。

对莱斯特郡消防与救援服务局开展的半结构式访谈显示，在紧急情况下，由于应急车辆的尺寸、重量和动力，车辆可能能够穿越约25 cm深的水（低于车轮拱罩）。一辆消防车重约12 T，驾驶员也受过穿越洪水的训练。该阈值深度也与先前研究一致（HR Wallingford，2006；Dawson 等，2011；Pregnolato 等，2016）。尽管高流速可能妨碍应急响应人员成功穿越洪水，但由于地形洼地积水，模型中的流速通常<1 m s-1。

因此，在评估网络限制位置时，选择水深作为主要因素。由此，地表水洪水情景的阈值水深设为25 cm，水深≥25 cm的区域被视为特定道路路段上应急车辆通行的限制。

随后对水深≥25 cm的地表水洪水区域进行处理，移除与 ITN 不重叠或不相交、因而不会用于分析的其他多边形（即在